Plant Acoustic Emissions - Analysis Report

Experiment Date: 2026-06-21
Analysis Date: 2026-06-21

Summary

This report presents the spectral analysis of acoustic emissions from a plant using three piezoelectric sensors with different configurations.

Channel Statistics

Channel	Configuration	Captures	Peak-to-Peak (mV)	RMS (mV)
CH1	Piezo + LM358 + 820kΩ (10:1)	20	1020.00	509.76
CH2	Piezo + 820kΩ (1:1)	20	28.40	5.19
CH3	Piezo direct (1:1)	20	9.20	2.00

Dominant Frequency Components

Channel 1

Rank	Frequency (Hz)	Amplitude (dB)	Prominence (dB)
1	50.0	-54.6	10.5
2	358.3	-46.9	5.0
3	1358.3	-41.3	4.2
4	3425.0	-35.2	4.0
5	2325.0	-38.2	4.0
6	2675.0	-37.3	4.0
7	150.0	-57.1	3.8
8	1575.0	-40.6	3.7
9	633.3	-47.0	3.4
10	1808.3	-41.0	3.0

Channel 2

Rank	Frequency (Hz)	Amplitude (dB)	Prominence (dB)
1	50.0	-56.7	29.0
2	150.0	-66.9	24.1
3	250.0	-75.4	16.1
4	100.0	-74.2	15.8
5	350.0	-80.7	10.7
6	200.0	-81.8	8.6
7	300.0	-82.4	8.2
8	450.0	-83.4	6.8
9	391.7	-84.5	5.4

Channel 3

Rank	Frequency (Hz)	Amplitude (dB)	Prominence (dB)
1	50.0	-64.6	33.1
2	150.0	-74.9	29.4
3	250.0	-79.1	24.7
4	450.0	-87.4	14.0
5	350.0	-92.0	10.0
6	650.0	-90.7	8.5
7	550.0	-93.8	5.5
8	100.0	-99.5	3.8
9	500.0	-95.3	3.4

Observations

1. **Signal Amplification:** CH1 (amplified) shows ~44× higher amplitude than CH2.
2. **Frequency Consistency:** Similar frequency components (2-5 kHz) appear across all channels.
3. **Mains Interference:** 50 Hz component detected in CH2 and CH3.
4. **Mechanical Origin:** Consistent frequencies across channels suggest genuine mechanical vibrations.

Visualizations

- Time Domain Comparison
- Frequency Domain Comparison
- Channel Summary Table